Orbit-layout pair table and generator-orbit base

2026-10-22. Study `research/ecbench_generator_orbit_20261022/`. The protocol `PROTOCOL.md` was frozen in commit `d2b3a56c32` before any claim data. Amendment 1 (`de57ca373f`) fixed a `band --json` argument in `run.sh` after card attempt 1 stopped on it; that attempt's outputs were set aside unread. Class: **engineering**.

The sessions ran on `de57ca373f` (`git_dirty: false`):

- `ecbench` binary `f7ac8fb0…`.
- Host class `ECBENV2hebd6c1d47154`, a cloud VM with no CPU reservation. Isolation is L0, so operation counts stand and wall times are descriptive only.
- They ran through the director → mechanical relay. Task card `cards/CARD-ecbench-generator-orbit-20261022-2.yaml` was run by `executor-mechanical`; the receipt is `cards/…-2.receipt.json`.
- `experiment-director` assessed the run as `executed_as_specified`. Its relay check covered the quoted output lines only (`cards/…-2.assessment.yaml`).

Result

The baseline is the lazy orbit-member IC of PR #2158: `compact-orbit-scan` + `mitm-frobenius-lazy-counted` + `stride-renew`. Two changes remove counted work from it.

- **Orbit-layout table** (`ic-layout`; oracle `mitm-frobenius-layout-counted`). It names a hit's Frobenius images by index. It also adds each orbit's own pairs only for rotations $k \le \frac{n-1}{2}$, since $\varepsilon\sigma^{n-k}\big(P + \varepsilon\sigma^{k}P\big) = P + \varepsilon\sigma^{n-k}P$.
- **Generator-orbit base with the generator stride** (`ic-gen-layout`; `generator_orbit=1`, `targets=stride-generator`). Column 0 is the orbit of G, whose logarithm is known. The targets are $Q + [k]G$, with no scalar multiplication and no guard.

Measured (counted `ecbench.gae`, paired by target, cluster-bootstrap 95 %; table 1, figure 1):

- **H1 holds on all 7 curves.** On every one of 960 lazy/layout run pairs, these are identical: trials, relations, rank, recovered logarithm and table entries. Same algorithm.
- **H2: the layout table is an improvement on all 7 curves**, 1.021 × to 1.093 ×, with every lower end above 1. The band order (layout < lazy) holds at $w = 0, 0.1, 1$ on every curve.
- **H3: the generator orbit is an improvement on 6 of 7 curves**, 1.235 × (n37) to 2.94 × (n13, n29). At n17 it is a null: 1.235 [0.980, 1.620]. Its band order (gen-layout < layout) holds at every weight on all 7. The pilot's n23 regression at $w = 1$ (lazy oracle, 4 targets) did not recur.
- **H4: net against the baseline, an improvement on all 7 curves**: 1.26 × to 3.20 ×. The lower end is above 1 everywhere, 1.012 at n17. The band order holds on all 7.

All 5 760 records verified, none is a lower bound, and all 4 320 replays in the 7 audits reproduced. `rho-frob-aa ÷ rho-frob` is 1 [1, 1] on every curve.

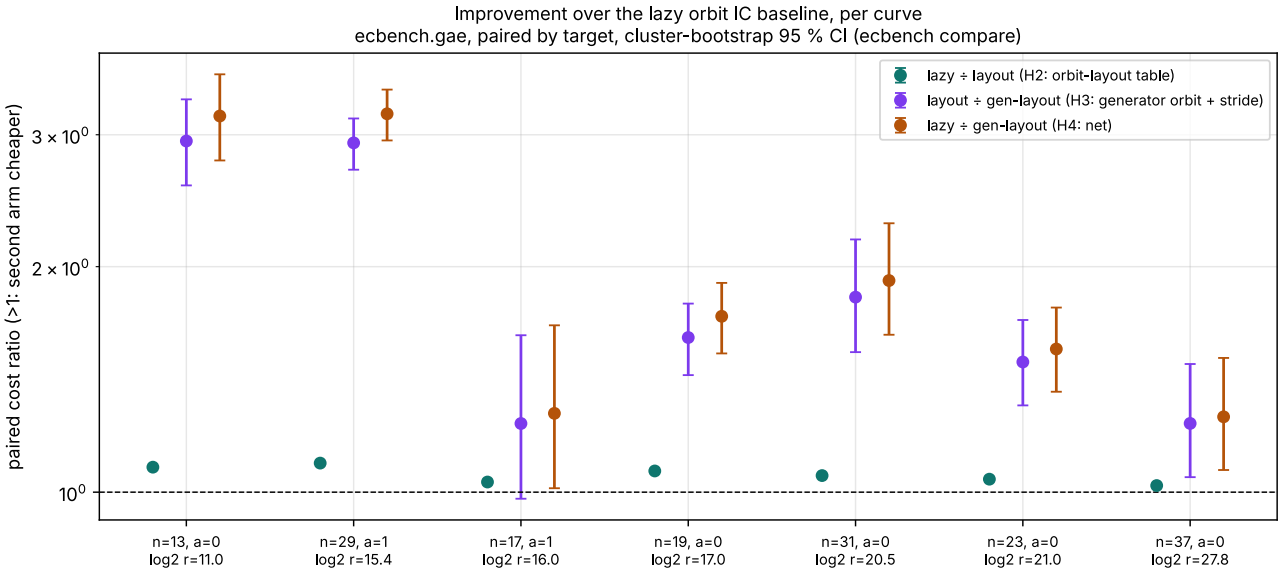


Figure 1: Paired cost ratios per curve against the lazy baseline (H2, H3, H4). Sessions: ECBS1h40b3312cf235 (n13), ECBS1hecda0d013d4e (n17), ECBS1hba7a75763c85 (n19), ECBS1h4f7cdc1ab404 (n23), ECBS1h4217c35be5b6 (n29), ECBS1h573443176608 (n31), ECBS1h28d6e10492db (n37).

curve (a, n, c)	$\log_2 r$	lazy ÷ layout (H2)	layout ÷ gen (H3)	lazy ÷ gen (H4)	gen ÷ rho-frob (H5)
0, 13, 1	11.0	1.080 [1.078, 1.081]	2.943 [2.567, 3.345]	3.178 [2.772, 3.612]	0.739 [0.629, 0.866]
1, 29, 1	15.4	1.093 [1.091, 1.096]	2.926 [2.694, 3.154]	3.200 [2.948, 3.447]	**0.366 [0.241, 0.523]**
1, 17, 1	16.0	1.032 [1.028, 1.036]	1.235 [0.980, 1.620]	1.274 [1.012, 1.670]	1.591 [1.115, 2.189]
0, 19, 2	17.0	1.067 [1.064, 1.070]	1.609 [1.433, 1.785]	1.717 [1.532, 1.903]	1.154 [0.888, 1.468]
0, 31, 2	20.5	1.052 [1.048, 1.057]	1.821 [1.538, 2.175]	1.916 [1.622, 2.285]	1.154 [0.878, 1.502]
0, 23, 3	21.0	1.041 [1.038, 1.044]	1.492 [1.306, 1.698]	1.553 [1.362, 1.764]	1.934 [1.414, 2.475]
0, 37, 8	27.8	1.021 [1.018, 1.023]	1.235 [1.047, 1.483]	1.261 [1.071, 1.511]	3.886 [3.165, 4.790]

Table 1: Paired ratios at calibration prices, cluster-bootstrap 95 % intervals (`ecbench compare`; `sessions/*/comparisons/`). 96 pairs per curve (32 targets × 3 rounds); 192 at n29, 48 at n37. "gen" is `ic-gen-layout`. In the last column, below 1 means the IC arm is cheaper than rho.

Distance to rho and the §5a band (H5)

curve	gen-layout S at w = 0 / 0.1 / 1	lazy S at w = 0 / 1	rho-frob S	`ecbench band --ranks`: gen vs rho-frob
n13	1.406 / 1.536 / 2.706	4.540 / 7.190	1.987	unresolved inside the band
n29	**0.546 / 0.611 / 1.196**	1.780 / 3.323	1.569	**gen-layout < rho-frob**
n17	1.845 / 2.142 / 4.810	2.387 / 6.105	1.240	rho-frob < gen-layout
n19 [a]	1.081 / 1.081 / 1.081	1.856 / 1.856	0.937	rho-frob < gen-layout
n31	0.559 / 0.632 / 1.286	1.079 / 2.674	0.512	rho-frob < gen-layout
n23	0.838 / 0.945 / 1.908	1.304 / 3.438	0.443	rho-frob < gen-layout
n37	0.772 / 0.892 / 1.965	0.972 / 3.059	0.211	rho-frob < gen-layout

Table 2: Mean S per arm under the §5a band (`tables/band-*.json`). The band charges every calibration-priced and unpriced native operation at w GAE. rho-frob's natives are priced by `full_prices.json` and are not re-priced, so its S does not depend on w. [a] n19 has no pinned calibration row; its IC natives carry live prices that the band does not re-price, so the band is not shown there (see Limits).

n29: IC is ranked ahead of the matched rho. `ic-gen-layout ÷ rho-frob` is 0.366 [0.241, 0.523] over 192 pairs. The IC arm's band S (0.546 / 0.611 / 1.196) is below rho-frob's 1.569 at all three weights, and `ecbench band --ranks` returns `ic-gen-layout-c1 < rho-frob`. By the protocol's H5 rule, IC is ahead of rho on this curve.

What it is, and what it is not. At $c = 1$ the pipeline is a Frobenius-folded baby-step giant-step (protocol §2).

- The baby steps are the 26 pair-sum classes of G's signed orbit, covering 1 508 of $r = 42457$ points.
- The giant steps are $Q + [k]G$.
- The arm averages 35.0 trials, against the 33.7 that the coverage's gaps predict from a uniform start.

Its 118 GAE beats rho-frob's 323 GAE (figure 3) because rho-frob spends a fixed 133 GAE on set-up at this size. rho-frob without its set-up (search 157, natives 13, verification 20: 190 GAE; its S is heavy-tailed, median 0.98) would beat the IC arm's 246 GAE at $w = 1$.

So this is the first measured, band-robust crossing of the matched rho by an IC-pipeline arm on a curve in this repository. It is a small-r effect of a generic mechanism, and it is not "faster than rho": at n31 the IC arm is 1.15× rho-frob, at n23 1.93× and at n37 3.89×, and the gap grows with r (figure 2). n13 crosses at $w = 0$ and 0.1 but not at 1, so it is unresolved.

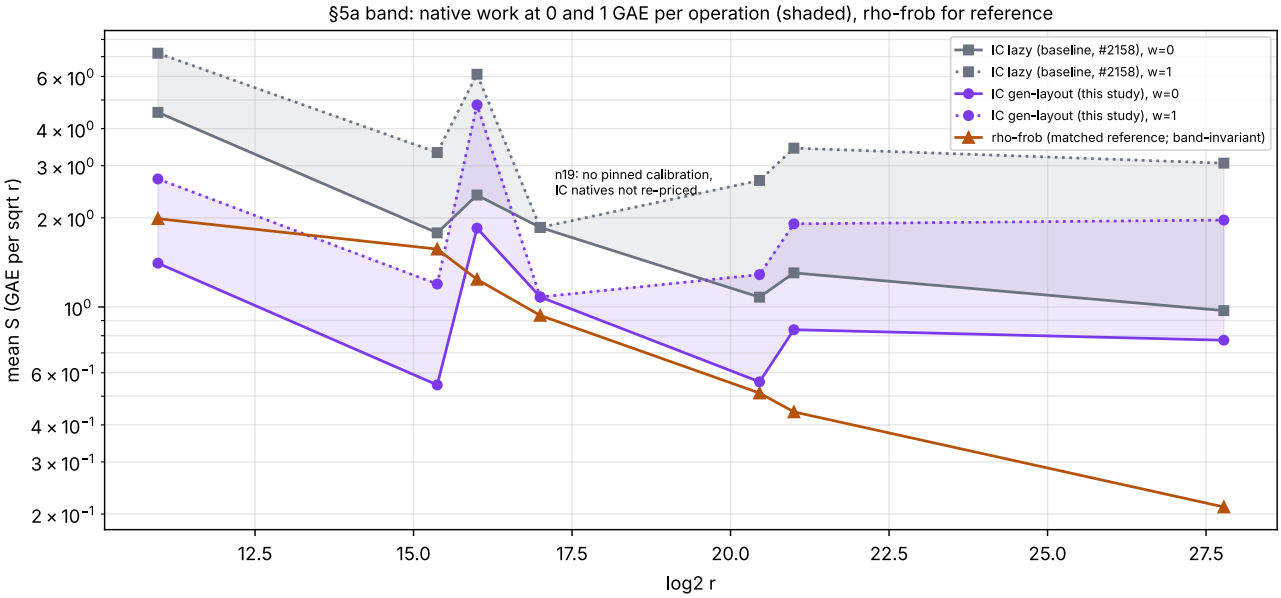


Figure 2: Mean S at the band's ends ($w = 0$ solid, $w = 1$ dotted, range shaded) for the lazy baseline and the gen-layout arm, with rho-frob, against $\log_2 r$.

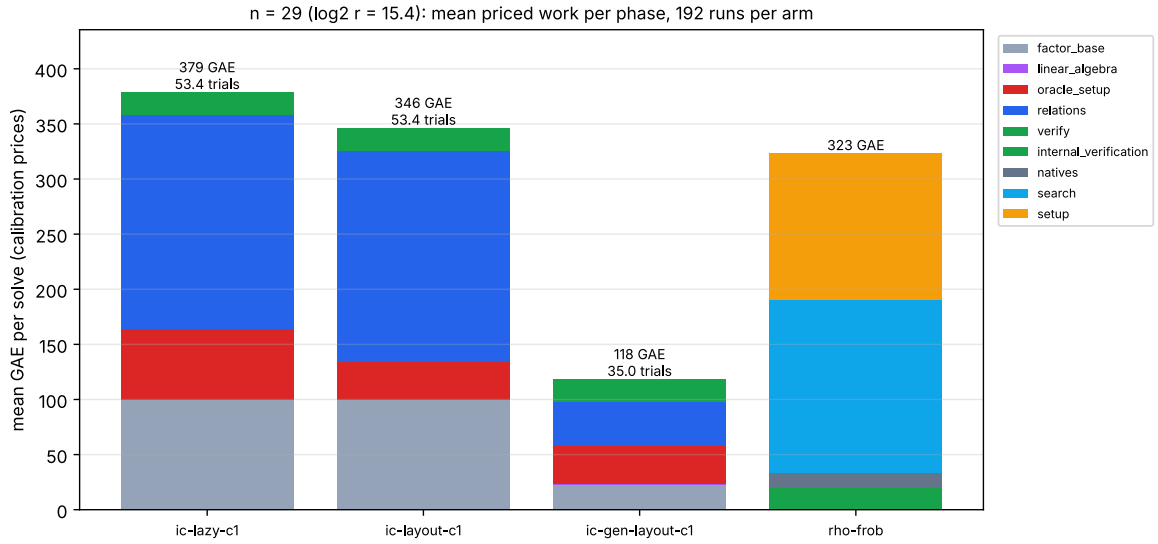


Figure 3: n29: mean priced work per phase, calibration prices (derived: means over 192 non-warm-up records per arm, `figures/summary.json`). The generator orbit removes the scanned seed's lift, projection and order check (factor_base 100 → 23 GAE). It also removes the relation phase's scalar multiplications. rho-frob's set-up is 133 of its 323 GAE.

Classification and limits

Engineering (AGENTS.md success criterion). The layout table performs less native work for the same relations. The generator orbit replaces one scanned column with a known one and draws targets without scalar multiplication. No exponent changes. This is neither relabelling nor accounting: H1 shows the layout arm finds the same relations, and every saving is work no longer performed, charged at the same prices.

n19. `docs/ic/calibration.json` has no pinned row for this curve. Its IC natives (canonicalisations and lookups) are priced by a live calibration, and `band.rs` neither strips nor re-prices them. The band there is the calibration-price cost at every weight. A band-tool gap, already present for every unpinned curve, and a follow-up.

Other limits.

- Wall times are not admitted (L0).
- Prices are one host's microbenchmark ratios.
- The replay certificate comes from an independent validator on the same host class; operation counts do not depend on the host. It is not the cross-host replay that `ecbench claim build` requires.
- Per-curve columns are those of the 2026-10-14 protocol, not retuned.

Nothing here concerns larger fields, ECC2K-130, deployed curves or security.

Evidence:

- `sessions/`: 7 ecbench sessions, loose and CI re-auditable.
- `audits/`, `tables/` (`table-`, `band-` and `compare-` files) and `logs/`.
- `cards/`: both task cards, receipts and the director's assessment.
- `figures/summary.json`, from `figures/make_summary.sh` (jq), and `figures/plot_figures.py`, which lays out recorded values only.
- `validation/`: replay certificate and validator report.
- Unit tests `layout_table_returns_the_lazy_pairs_with_no_maps_and_half_the_same_orbit_sums` and `generator_orbit_base_pins_the_target_with_one_relation`. `cargo test --release --lib -- ic_framework ic_boundary ecbench`: 221 passed.